THE CLEBSCH AND SCHLÄFLI GRAPHS

Abstract. A strongly regular graph with parameters $(16,5,0,2)$ is isomorphic to the folded 5-cube, and so one with parameters $(16,10,6,6)$ is isomorphic to the Clebsch graph; a strongly regular graph with parameters $(27,16,10,8)$ is isomorphic to the Schläfli graph. These are prerequisites SE2 and SE3 of `cgss/prerequisites.md`, used in step ⟨1⟩11 of Remark 4.14 of [CGSS]. We give a direct proof of the first, reduce the second to the uniqueness assertion of Lemma 2.10 of [CGSS] as repaired in the blueprint, check both with SageMath, and describe the shape of the Lean statements.

1. Statement

All graphs are simple and finite. For a vertex v of a graph X we will write $N(v)$ for its neighbourhood, $u \sim w$ when u and w are adjacent, and $\overline{X}$ for the complement of X. A graph X on n vertices is *strongly regular with parameters* (n,k,a,c) if it is k-regular, any two adjacent vertices have exactly a common neighbours, and any two distinct nonadjacent vertices have exactly c common neighbours. This is Mathlib's `SimpleGraph.IsSRGWith n k a c`, with no further condition; in particular X may be complete or empty. For sets s,t we will write $s \triangle t$ for the symmetric difference, and for a set Ω we will write $\binom{\Omega}{2}$ for the set of its 2-subsets.

Definition 1.1. Let $\Omega_5 = \{1,\dots,5\}$, and let V_{16} be the set of the subsets of Ω_5 of even size; thus V_{16} consists of $\emptyset$, the ten 2-subsets and the five 4-subsets, and $|V_{16}| = 16$. The *folded 5-cube* $\square_5$ is the graph on V_{16} in which $s \sim t$ if and only if $|s \triangle t| = 4$. The *Clebsch graph* is its complement $\overline{\square_5}$.

For distinct $s,t \in V_{16}$ the size $|s \triangle t| = |s| + |t| - 2|s \cap t|$ is even, positive and at most 5, so it is 2 or 4. Thus s and t are adjacent in the Clebsch graph if and only if $|s \triangle t| = 2$, and the Clebsch graph is the halved 5-cube. The folded 5-cube is the 5-cube with antipodal vertices identified: each pair $\{x, \Omega_5 \setminus x\}$ contains exactly one set of even size, and $x \mapsto x \cup \{i\}$ or $x \setminus \{i\}$ becomes $s \mapsto \Omega_5 \setminus (s \triangle \{i\})$, at distance 4 from s. Neither description is needed below; both are checked in Section 5. We follow [CGSS] Example 4.8 and `cgss/notation.md` §6 in giving the name to the 10-regular graph; SageMath's `graphs.ClebschGraph()` is the 5-regular one, our $\square_5$. [CGSS] Example 4.8 gives the Clebsch graph by the 16 vectors $\mathbf{e}_i + \mathbf{e}_5$, $\mathbf{e}_i + \mathbf{e}_6$, $\mathbf{c} - \mathbf{e}_i - \mathbf{e}_7$ and $\mathbf{c} - \mathbf{e}_i - \mathbf{e}_8$ $(1 \le i \le 4)$ of E_8, with $\mathbf{c} = \frac{1}{2}\sum_{m=1}^{8} \mathbf{e}_m$, two of them adjacent when their inner product is 1; this graph is isomorphic to $\overline{\square_5}$ (Section 5), and in Lean the two are different graphs joined by an explicit isomorphism (Section 6).

Definition 1.2. Let $\Omega_6 = \{1,\dots,6\}$, and let V_{27} be the set of the 27 symbols a_i, b_i $(i \in \Omega_6)$ and $c_{ij} = c_{ji}$ $(\{i,j\} \in \binom{\Omega_6}{2})$. Let Q be the graph on V_{27} with the following edges, and no others:

(1) $a_i \sim b_j$ for $i \ne j$;
(2) $a_i \sim c_{jk}$ and $b_i \sim c_{jk}$ for $i \in \{j,k\}$;
(3) $c_{ij} \sim c_{kl}$ for $\{i,j\} \cap \{k,l\} = \emptyset$.

The *Schläfli graph* is its complement $\overline{Q}$. Thus in $\overline{Q}$ we have $a_i \sim a_j$ and $b_i \sim b_j$ for $i \ne j$, $a_i \sim b_i$, $a_i \sim c_{jk}$ and $b_i \sim c_{jk}$ for $i \notin \{j,k\}$, and $c_{ij} \sim c_{kl}$ when $|\{i,j\} \cap \{k,l\}| = 1$.

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This is Schläfli's double six: a_i, b_i and c_{ij} are the 27 lines on a smooth cubic surface, Q is the graph of meeting lines, and the Schläfli graph is the graph of skew lines. The triangles of Q are the 45 tritangent planes, and Q is the collinearity graph of the generalized quadrangle of order $(2,4)$. We will not use this. [BCN] 3.11 (ii) gives the same graph in coordinates: $a_i = \mathbf{e}_i + \mathbf{e}_7$, $b_i = \mathbf{e}_i + \mathbf{e}_8$ and $c_{ij} = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_j$ in E_8, with $\mathbf{c} = \frac{1}{2}\sum_{m=1}^{8} \mathbf{e}_m$, and two vertices adjacent in $\overline{Q}$ when their inner product is 1 (it is 0 otherwise); this too is checked in Section 5.

[CGSS] Example 4.6, with $\mathbf{c}$ in place of the misprinted $\frac{1}{3}\sum_k \mathbf{e}_k$ (`cgss/errata.md`, Error 2), represents the Schläfli graph by other vectors of E_8: $\mathbf{x}_{ij} = \mathbf{e}_i + \mathbf{e}_j$ for $1 \le i < j \le 6$, and $\mathbf{y}_i = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_7$ and $\mathbf{z}_i = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_8$ for $1 \le i \le 6$, two of them adjacent when their inner product is 1. The bijection

$$a_i \mapsto \mathbf{y}_i = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_7, \qquad b_i \mapsto \mathbf{z}_i = \mathbf{c} - \mathbf{e}_i - \mathbf{e}_8, \qquad c_{ij} \mapsto \mathbf{x}_{ij} = \mathbf{e}_i + \mathbf{e}_j$$

is an isomorphism from $\overline{Q}$ onto the graph of these vectors: for $i \ne k$ we have $(\mathbf{y}_i, \mathbf{y}_k) = (\mathbf{z}_i, \mathbf{z}_k) = 1$; $(\mathbf{y}_i, \mathbf{z}_k)$ is 1 if $i = k$ and 0 otherwise; $(\mathbf{x}_{ij}, \mathbf{y}_k) = (\mathbf{x}_{ij}, \mathbf{z}_k)$ is 1 if $k \notin \{i, j\}$ and 0 otherwise; and $(\mathbf{x}_{ij}, \mathbf{x}_{kl}) = |\{i,j\} \cap \{k,l\}|$ (the blueprint, Example 4.6, step `s46ip`). These are the adjacencies of $\overline{Q}$ listed in Definition 1.2. The vectors of [BCN] and of [CGSS] are different, but under the two labellings they have the same Gram matrix. The correspondence is checked in Section 5, and it is the certificate isomorphism of Section 6.

Why this model. Of the three usual descriptions (the 27 lines, the 27 vectors of E_8 above, and a quadric or Hermitian model over a finite field), the first is the easiest to state in Lean: its vertex type is a sum of two finite types, its adjacency is decidable by cases on four set conditions, and it needs no inner products and no field. Its vertices correspond one to one to the vectors of the second and to those of [CGSS] Example 4.6, so that the representation of the Schläfli graph in E_8 becomes a check on explicit vectors of the form of step $\langle 1 \rangle 9$ of Remark 4.14 of [CGSS]; and it uses 2-subsets as the folded 5-cube does.

Theorem 1.3 (SE2)**.** *Let X be a strongly regular graph with parameters $(16, 5, 0, 2)$. Then X is isomorphic to the folded 5-cube $\square_5$. Thus a strongly regular graph with parameters $(16, 10, 6, 6)$ is isomorphic to the Clebsch graph.*

Theorem 1.4 (SE3)**.** *Let X be a strongly regular graph with parameters $(27, 16, 10, 8)$. Then X is isomorphic to the Schläfli graph $\overline{Q}$. Equivalently, a strongly regular graph with parameters $(27, 10, 1, 5)$ is isomorphic to Q.*

2. Complements

Lemma 2.1. *If X is strongly regular with parameters (n, k, a, c), then $\overline{X}$ is strongly regular with parameters $(n,\ n-k-1,\ n-2-2k+c,\ n-2k+a)$, read in $\mathbb{Z}$.*

A negative parameter here belongs to a condition that is vacuous: $n-2-2k+c$ is the number of common neighbours in $\overline{X}$ of two vertices adjacent in $\overline{X}$, and $n-2k+a$ that of two distinct vertices nonadjacent in $\overline{X}$, so either is negative only when there is no such pair, as when X is complete or empty.

Proof. Each vertex has $n-1-k$ neighbours in $\overline{X}$. Let $u \ne w$. A common neighbour of u and w in $\overline{X}$ is a vertex outside $\{u, w\} \cup N(u) \cup N(w)$. If $u \sim w$ in X, then $u, w \in N(u) \cup N(w)$ and $|N(u) \cup N(w)| = 2k - a$, so u and w, which are nonadjacent in $\overline{X}$, have $n - 2k + a$ common neighbours there. If $u \not\sim w$ in X, then $u, w \notin N(u) \cup N(w)$ and $|N(u) \cup N(w)| = 2k - c$, so u and w, which are adjacent in $\overline{X}$, have $n - 2 - 2k + c$ common neighbours there. □

This is `SimpleGraph.IsSRGWith.compl`, in Mathlib's file `Mathlib/Combinatorics/SimpleGraph/StronglyRegular.lean`, which states the parameters as

`n (n - k - 1) (n - (2 * k - μ) - 2) (n - (2 * k - ℓ))`

in truncated subtraction. The two forms agree whenever the condition is not vacuous, and for the four parameter sets below no subtraction is truncated (the smallest intermediate value is $n-(2k-c)-2=0$, for $(16,10,6,6)$). The lemma turns $(16,5,0,2)$ and $(16,10,6,6)$ into each other, and $(27,16,10,8)$ and $(27,10,1,5)$ into each other. Since $\overline{\overline{X}}=X$, and a bijection that preserves adjacency and nonadjacency of distinct vertices of X and Y does the same for $\overline{X}$ and $\overline{Y}$, we have $X\cong Y$ if and only if $\overline{X}\cong\overline{Y}$.

3. The folded 5-cube

Lemma 3.1. *The folded 5-cube is strongly regular with parameters* $(16,5,0,2)$*, and the Clebsch graph with parameters* $(16,10,6,6)$*.*

Proof. For $p\in V_{16}$ the map $s\mapsto s\triangle p$ is a permutation of V_{16} with $(s\triangle p)\triangle(t\triangle p)=s\triangle t$, and so an automorphism of $\square_5$; it sends p to $\emptyset$. Thus it suffices to count at the vertex $\emptyset$. Its neighbours are the five 4-subsets $\Omega_5\setminus\{i\}$. Two of them have symmetric difference of size 2, so they are not adjacent, and every edge at $\emptyset$ lies in no triangle; thus $a=0$. A vertex not adjacent to $\emptyset$ and distinct from it is a 2-subset P, and $|(\Omega_5\setminus\{i\})\triangle P|$ is 4 if $i\in P$ and 2 if $i\notin P$; so $\emptyset$ and P have exactly the two common neighbours $\Omega_5\setminus\{i\}$, $i\in P$, and $c=2$. The second assertion follows by Lemma 2.1. □

Proof of Theorem 1.3. Let X be strongly regular with parameters $(16,5,0,2)$, fix a vertex u, and put $N=N(u)$ and $R=V(X)\setminus(\{u\}\cup N)$. Then $|N|=5$ and $|R|=10$.

Step 1: N is independent. If $x,y\in N$ were adjacent, then u would be a common neighbour of the adjacent vertices x and y, whereas $a=0$.

Step 2: the pairs. Let $w\in R$. Then $w\neq u$ and $w\not\sim u$, so w and u have exactly $c=2$ common neighbours, which are the neighbours of w in N. We will write $\varphi(w)=N(w)\cap N\in\binom{N}{2}$.

Step 3: φ is a bijection $R\to\binom{N}{2}$. Suppose that $w\neq w'$ in R and $\varphi(w)=\varphi(w')=\{x,y\}$. Then $x\neq y$, and $x\not\sim y$ by Step 1, while u, w and w' are three distinct common neighbours of x and y; this contradicts $c=2$. Thus φ is injective, and since $|R|=10=\binom{5}{2}=\left|\binom{N}{2}\right|$, it is a bijection.

Step 4: adjacency in R is disjointness. Let $w\in R$, with $\varphi(w)=\{x,y\}$. The five neighbours of w are x, y and three vertices of R, since $w\not\sim u$. Let $w'\in R$ be a neighbour of w. If $x\in\varphi(w')$, then x would be a common neighbour of the adjacent vertices w and w', whereas $a=0$; thus $x\notin\varphi(w')$, and likewise $y\notin\varphi(w')$. Thus $\varphi(w')$ is one of the three 2-subsets of the 3-set $N\setminus\{x,y\}$. By Step 3, the three neighbours of w in R have three distinct such pairs, so they are exactly the three vertices $w'\in R$ with $\varphi(w')\cap\varphi(w)=\emptyset$. Thus for $w\neq w'$ in R we have $w\sim w'$ if and only if $\varphi(w)\cap\varphi(w')=\emptyset$.

Step 5: the isomorphism. Choose a bijection $\beta\colon N\to\Omega_5$, and define $f\colon V(X)\to V_{16}$ by

$$f(u)=\emptyset,\qquad f(x)=\Omega_5\setminus\{\beta(x)\}\ (x\in N),\qquad f(w)=\beta(\varphi(w))\ (w\in R).$$

By Step 3, f is a bijection onto $\{\emptyset\}\cup\{\text{4-subsets}\}\cup\binom{\Omega_5}{2}=V_{16}$. We compare adjacency for distinct vertices p,q, by the parts that contain them.

(1) u and $x\in N$: adjacent in X, and $|\emptyset\triangle(\Omega_5\setminus\{\beta(x)\})|=4$.
(2) u and $w\in R$: not adjacent, and $|\emptyset\triangle f(w)|=2$.
(3) $x,y\in N$: not adjacent by Step 1, and $|f(x)\triangle f(y)|=|\{\beta(x),\beta(y)\}|=2$.
(4) $x\in N$ and $w\in R$: $x\sim w$ if and only if $x\in\varphi(w)$, that is, $\beta(x)\in f(w)$; and, as in the proof of Lemma 3.1, $|f(x)\triangle f(w)|=4$ if and only if $\beta(x)\in f(w)$.
(5) $w,w'\in R$: $w\sim w'$ if and only if $f(w)\cap f(w')=\emptyset$ by Step 4, and for two distinct 2-subsets $|f(w)\triangle f(w')|=4-2|f(w)\cap f(w')|$ is 4 exactly when they are disjoint.

Thus f is an isomorphism $X \to \square_5$.

For the consequence, let Y be strongly regular with parameters $(16, 10, 6, 6)$. By Lemma 2.1, $\overline{Y}$ is strongly regular with parameters $(16, 5, 0, 2)$, so $\overline{Y} \cong \square_5$, and $Y \cong \overline{\square_5}$ by Section 2. □

Remark 3.2. The proof uses $c = 2$ twice (Steps 2 and 3), $a = 0$ twice (Steps 1 and 4), and the valency 5 and $n = 16$ in Steps 3 and 4 (through $|N| = 5$, $|R| = 10$ and the three neighbours of w in R). The remaining instances of $c = 2$ are then consequences. For $w \in R$ and $z \in N \setminus \varphi(w)$, the common neighbours of w and z are the two vertices $w' \in R$ with $\varphi(w') = \{z, z'\}$, where $\{z, z'\} \subseteq N \setminus \varphi(w)$. For distinct nonadjacent $w, w' \in R$, whose pairs meet by Step 4 in one vertex x, the common neighbours of w and w' are x and the vertex $w'' \in R$ whose pair is $N \setminus (\varphi(w) \cup \varphi(w'))$. Thus the argument does not need them. The subgraph induced on R is the Petersen graph (the Kneser graph of the 2-subsets of a 5-set), in agreement with [BvM] §10.7; we do not use SE1.

4. The Schläfli graph

We will use the predicate of the blueprint (`blueprint/sec2-lines.tex`, Definition *the triangle property*, label `def:triangle-property`): a graph Y has the *triangle property* if every edge xy has exactly one common neighbour z, and every vertex outside $\{x, y, z\}$ is adjacent to exactly one of x, y, z.

Lemma 4.1. *Let Y be strongly regular with parameters $(n, k, 1, c)$, where $n + 3 = 3k$. Then Y has the triangle property.*

The condition $n+3 = 3k$ is $n = 3k-3$, and $n-3 = 3(k-2)$ as in the blueprint (Remark 4.14, `lem:414-GQ1`), written without subtraction, and so in the form of the Lean hypothesis (Section 6).

Proof. Let $x \sim y$. Since $a = 1$, the edge xy has exactly one common neighbour z, and $T = \{x, y, z\}$ is a triangle. A vertex $w \notin T$ is adjacent to at most one vertex of T: if it were adjacent to two, say to x and y (the other cases are the same, since xz and yz are also edges whose unique common neighbour is the third vertex of T), then w and z would be two distinct common neighbours of x and y. Each vertex of T has exactly two neighbours in T, and so $k - 2$ neighbours outside T. Thus there are $3(k - 2) = n - 3$ edges between T and $V(Y) \setminus T$, and no two of them end at the same vertex of $V(Y) \setminus T$. Since $|V(Y) \setminus T| = n - 3$, every vertex outside T is the end of exactly one of them, that is, adjacent to exactly one vertex of T. □

The condition $n + 3 = 3k$ holds for $(9, 4, 1, 2)$, $(15, 6, 1, 3)$ and $(27, 10, 1, 5)$, and more generally for the collinearity graph of a generalized quadrangle of order $(2, t)$, with $n = 3(2t + 1)$ and $k = 2(t + 1)$; the lemma is the converse half of GQ1. The parameter c is not used.

Lemma 4.2. *The graph Q is strongly regular with parameters $(27, 10, 1, 5)$, and the Schläfli graph with parameters $(27, 16, 10, 8)$.*

Proof. The symmetric group on Ω_6, acting on the indices, and the exchange $a_i \leftrightarrow b_i$ are automorphisms of Q. Under the group they generate, every vertex lies in the orbit of a_1 or of c_{12}, every edge in the orbit of a_1b_2, a_1c_{12} or $c_{12}c_{34}$, and every pair of distinct nonadjacent vertices in the orbit of a_1a_2, a_1b_1, a_1c_{23} or $c_{12}c_{13}$. The vertex a_1 has the ten neighbours $b_2, \dots, b_6$ and $c_{12}, \dots, c_{16}$, and c_{12} has the ten neighbours a_1, a_2, b_1, b_2 and the six c_{kl} with $\{k, l\} \subseteq \{3, 4, 5, 6\}$. The common neighbours are:

pair	common neighbours	number
a_1b_2	c_{12}	1
a_1c_{12}	b_2	1
$c_{12}c_{34}$	c_{56}	1
a_1a_2	$b_3, b_4, b_5, b_6,\ c_{12}$	5
a_1b_1	$c_{12}, \dots, c_{16}$	5
a_1c_{23}	$b_2,\ b_3,\ c_{14}, c_{15}, c_{16}$	5
$c_{12}c_{13}$	$a_1,\ b_1,\ c_{45}, c_{46}, c_{56}$	5

The second assertion follows by Lemma 2.1. □

We will use the following result of the blueprint, which the repaired Lemma 2.10 of [CGSS] applies in its step `210unique`.

Proposition 4.3 (`blueprint/sec2-lines.tex`, Lemma *isomorphism from the order*, label `lem:triangle-iso`)**.** *Let Y and Z be finite graphs with the triangle property, each with an edge, neither with a vertex adjacent to every other vertex, and with $|V(Y)| = |V(Z)|$. Then $Y \cong Z$.*

This is the graph-theoretic uniqueness assertion behind Lemma 2.10 of [CGSS], stated without a vector representation; its Lean name is

`SimpleGraph.HasTriangleProperty.nonempty_iso_of_card_eq`

(`cgss/notation.md` §10, row Lemma 2.10). We do not use the form

`LineSystem.Hyp24.orthGraph_iso_of_card_Γa_eq`

of the uniqueness assertion, which asks that both graphs be orthogonality graphs of instances of Hypothesis 2.4; nor do we use the classification of the generalized quadrangles with lines of size 3 (GQ2).

Proof of Theorem 1.4. Let X be strongly regular with parameters $(27, 16, 10, 8)$. By Lemma 2.1, $\overline{X}$ is strongly regular with parameters $(27, 10, 1, 5)$, and so is Q by Lemma 4.2. Since $27+3 = 3\cdot 10$, both have the triangle property by Lemma 4.1. Both have an edge, since they are 10-regular on a nonempty set, and neither has a vertex adjacent to every other vertex, since $10 < 26$. They have 27 vertices each. By Proposition 4.3, $\overline{X} \cong Q$, and so $X \cong \overline{Q}$ by Section 2. The equivalent form is the same argument without the first complementation. □

Thus SE3 in Lean is a reduction to the result of Section 2 of the blueprint: it needs Mathlib's `SimpleGraph.IsSRGWith.compl`, Lemma 4.1, the parameters of Q, and `SimpleGraph.HasTriangleProperty.nonempty_iso_of_card_eq`. Lemma 4.1 also shows that the strongly regular graphs with parameters $(9, 4, 1, 2)$ and $(15, 6, 1, 3)$ are unique, by the same argument with $L_2(3)$ and the complement of $T(6)$ in place of Q (the orthogonality graphs of E_6 and E_7 in Section 2 of the blueprint; `gq21` and `gq22` in `cgss/notation.md` §6).

5. The computation

The script `cgss/proofs/clebsch/checks/clebsch_check.sage`, run with SageMath 10, does the following.

(1) It builds $\square_5$ on the even subsets of $\{0, \dots, 4\}$ and checks that it is strongly regular with parameters $(16, 5, 0, 2)$, that the Clebsch graph has parameters $(16, 10, 6, 6)$ and is the graph of symmetric differences of size 2, that $\square_5$ is isomorphic to `graphs.ClebschGraph()`, to the antipodal quotient of `graphs.CubeGraph(5)`, and that the Clebsch graph is isomorphic to the halved 5-cube.

(2) At each of the 16 vertices u of `graphs.ClebschGraph()` it checks Steps 1 to 4 of the proof of Theorem 1.3 ($N(u)$ independent, every $w \in R$ with two neighbours in $N(u)$, φ injective, three neighbours in R, adjacency in R equal to disjointness of the pairs), and that the map f of Step 5 is an isomorphism onto $\square_5$.
(3) It builds Q and $\overline{Q}$ from the case lists of Definition 1.2, checks that the two lists describe complementary graphs, that the parameters are $(27, 10, 1, 5)$ and $(27, 16, 10, 8)$, that $\overline{Q}$ is isomorphic to `graphs.SchlaefliGraph()`, and that $\overline{Q}$ is the graph of the E_8 vectors of [BCN] 3.11 (ii) at inner product 1, and that the correspondence of Section 1 is an isomorphism from $\overline{Q}$ onto the graph of the vectors of [CGSS] Example 4.6 at inner product 1.
(4) It checks the formula of Lemma 2.1 against Mathlib's truncated one for the four parameter sets; the table of the proof of Lemma 4.2, pair by pair; that the permutations of the indices and the exchange $a_i \leftrightarrow b_i$ are automorphisms of Q, and that under the group they generate the vertices, the edges and the pairs of distinct nonadjacent vertices fall into 2, 3 and 4 orbits, with the representatives of that proof.
(5) It checks that the graph of the 16 vectors of [CGSS] Example 4.8 at inner product 1 is the subgraph of the graph of Example 4.6 induced on the neighbours of $\mathbf{e}_5 + \mathbf{e}_6$, and that the map of Section 6 is an isomorphism from it onto $\overline{\square_5}$, the certificate for Computation C9.
(6) It checks the triangle property for Q and for the complement of `graphs.SchlaefliGraph()`, the identity $n + 3 = 3k$ for the three parameter sets of Lemma 2.10 of [CGSS], that Q has 45 triangles with 5 through each vertex, and that its automorphism group has order $51840 = |W(\mathrm{E}_6)|$.

All checks pass. The computation is an independent check and not a part of the proof, which is complete without it.

6. The shape of the Lean statements

We will follow `cgss/notation.md` §10: the namespace `SimpleGraph`, lowerCamelCase for definitions, snake_case theorems naming their conclusion, theorems about strongly regular graphs in the namespace `SimpleGraph.IsSRGWith` so that dot notation works, and `nonempty_iso` at the start of a theorem that an isomorphism exists (rule 1). Thus the uniqueness theorems are `IsSRGWith.nonempty_iso_clebsch` and not `iso_clebsch`, which by rule 1 would be the name of a definition building the isomorphism.

The graph names are those of §6, with one definition for each name. The graphs `SimpleGraph.foldedFiveCube` and `SimpleGraph.gq24` are the combinatorial graphs $\square_5$ and Q of Definitions 1.1 and 1.2. The graphs `SimpleGraph.clebsch` and `SimpleGraph.schlafli` remain the graphs of the integer vectors of [CGSS] Examples 4.8 and 4.6, as in Computation C9 of the blueprint (`blueprint/sec4-leasteigenvalue.tex`, `comp:C9`), which gives them the parameters $(16, 10, 6, 6)$ and $(27, 16, 10, 8)$ (`clebsch_isSRGWith`, `schlafli_isSRGWith`). We do not define `clebsch := foldedFiveCube`^{c} or `schlafli := gq24`^{c}. Instead C9 supplies, as certificates checked by `decide`, the explicit isomorphisms

`def SimpleGraph.clebsch.foldedFiveCubeComplIso : clebsch ≃g foldedFiveCube`^{c}
`def SimpleGraph.schlafli.gq24ComplIso : schlafli ≃g gq24`^{c}

named as `LineSystem.E`$_6$`.latticeGraphIso` and `LineSystem.E`$_7$`.triangularGraphComplIso` of Computation C4 are: a definition that builds an isomorphism, in the namespace of the graph it starts from, named after the other graph (rule 1). The second is the inverse of the correspondence $a_i \mapsto \mathbf{y}_i$, $b_i \mapsto \mathbf{z}_i$, $c_{ij} \mapsto \mathbf{x}_{ij}$ of Section 1. The first, found by the script of

Section 5, sends

$\mathbf{e}_i+\mathbf{e}_5 \mapsto s_i$, $\mathbf{e}_i+\mathbf{e}_6 \mapsto s_i \triangle \{1,2\}$, $\mathbf{c}-\mathbf{e}_i-\mathbf{e}_7 \mapsto s_i \triangle \{1,3,4,5\}$, $\mathbf{c}-\mathbf{e}_i-\mathbf{e}_8 \mapsto s_i \triangle \{2,3,4,5\}$

for $1 \le i \le 4$, where $s_1 = \emptyset$, $s_2 = \{3,4\}$, $s_3 = \{3,5\}$ and $s_4 = \{4,5\}$.

object	Lean
$\square_5$	`SimpleGraph.foldedFiveCube : SimpleGraph S`, where `S = {s : Finset (Fin 5) // Even s.card}`, with `Adj s t := (s.1 △ t.1).card = 4`
$\overline{\square_5}$	`foldedFiveCube`^{c}, with `SimpleGraph.compl_foldedFiveCube_adj_iff` ($\lvert s \triangle t\rvert = 2$)
Q	`SimpleGraph.gq24 : SimpleGraph (Fin 2 × Fin 6 ⊕ T)`, where `T = {s : Finset (Fin 6) // s.card = 2}`
$\overline{Q}$	`gq24`^{c}, with `SimpleGraph.compl_gq24_adj_iff`
Clebsch graph (Ex. 4.8)	`SimpleGraph.clebsch` (C9), with `clebsch.foldedFiveCubeComplIso`
Schläfli graph (Ex. 4.6)	`SimpleGraph.schlafli` (C9), with `schlafli.gq24ComplIso`
Lemma 3.1	`SimpleGraph.foldedFiveCube_isSRGWith`, and `.compl`
Lemma 4.2	`SimpleGraph.gq24_isSRGWith`, and `.compl` for $\overline{Q}$
complements	`SimpleGraph.Iso.compl (f : X ≃g Y) : X`^{c} `≃g Y`^{c}
Lemma 4.1	`SimpleGraph.IsSRGWith.hasTriangleProperty`

The lemmas `compl_foldedFiveCube_adj_iff` and `compl_gq24_adj_iff` state the adjacencies of the complements given after Definitions 1.1 and 1.2; they replace the earlier proposals `clebsch_adj_iff` and `schlafli_adj_iff`, which named the complements when these were called `clebsch` and `schlafli`. In `gq24` the vertex $(0,i)$ is a_{i+1}, $(1,i)$ is b_{i+1}, and a 2-subset $\{i,j\}$ is $c_{i+1,j+1}$; adjacency is by cases: $(e,i) \sim (e',j)$ iff $e \ne e'$ and $i \ne j$, $(e,i) \sim s$ iff $i \in s$, and $s \sim t$ iff `Disjoint s.1 t.1`. Both vertex types have decidable equality and adjacency, and their cardinalities, 16 and 27, follow by `decide` or `simp`. The docstrings record the parameters, the names of the other sources (§6), and the descriptions of Section 1. With `X : SimpleGraph V`, `[Fintype V] [DecidableEq V] [DecidableRel X.Adj]`, the theorems are

```
theorem SimpleGraph.IsSRGWith.nonempty_iso_foldedFiveCube
    (h : X.IsSRGWith 16 5 0 2) : Nonempty (X ≃g foldedFiveCube)
theorem SimpleGraph.IsSRGWith.nonempty_iso_clebsch
    (h : X.IsSRGWith 16 10 6 6) : Nonempty (X ≃g clebsch)
theorem SimpleGraph.IsSRGWith.hasTriangleProperty
    (h : X.IsSRGWith n k 1 c) (hn : n + 3 = 3 * k) : X.HasTriangleProperty
theorem SimpleGraph.IsSRGWith.nonempty_iso_gq24
    (h : X.IsSRGWith 27 10 1 5) : Nonempty (X ≃g gq24)
theorem SimpleGraph.IsSRGWith.nonempty_iso_schlafli
    (h : X.IsSRGWith 27 16 10 8) : Nonempty (X ≃g schlafli)
```

(the hypothesis `hn` is $n+3 = 3k$ of Lemma 4.1, without truncated subtraction). The proof of `nonempty_iso_foldedFiveCube` defines the map f of Step 5 from a vertex `u`, the pair map `w` $\mapsto$ `X.neighborFinset u ∩ X.neighborFinset w` and an equivalence `X.neighborFinset u ≃ Fin 5` (from the cardinality 5 and `Finset.equivFinOfCardEq`); Steps 1 to 4 become four lemmas, and Step 5 a case split into the five cases. The proofs of `nonempty_iso_clebsch` and `nonempty_iso_schlafli` apply `IsSRGWith.compl`, then `nonempty_iso_foldedFiveCube` or `nonempty_iso_gq24`, then `Iso.compl` and `compl_compl`, which give `X ≃g foldedFiveCube`^{c} or `X ≃g gq24`^{c}, and compose with the inverse of the certificate `clebsch.foldedFiveCubeComplIso` or `schlafli.gq24ComplIso`. The proof of `nonempty_iso_gq24` applies

`SimpleGraph.HasTriangleProperty.nonempty_iso_of_card_eq` (Section 2 of the blueprint) to `X` and `gq24`, with `hasTriangleProperty` for both, the edge from the valency, and the absence of a universal vertex (Mathlib's `SimpleGraph.IsUniversal`) from $10 < 26$. The parameters of the models may be proved by `decide` (a check over at most 27^3 triples), or by the orbit argument of Lemmas 3.1 and 4.2 if the kernel is too slow.
The Mathlib declarations that we use are in Mathlib at commit `082e2d37` (4 September 2026), the version pinned in `lake-manifest.json`:

- `Mathlib/Combinatorics/SimpleGraph/StronglyRegular.lean`: the structure `SimpleGraph.IsSRGWith` with the fields `card`, `regular`, `of_adj`, `of_not_adj`; `SimpleGraph.IsSRGWith.compl`; `IsSRGWith.card_neighborFinset_union_of_adj` and `IsSRGWith.card_neighborFinset_union_of_not_adj` (the counts of Lemma 2.1).
- `Mathlib/Combinatorics/SimpleGraph/Basic.lean`: `SimpleGraph.compl_adj`, `SimpleGraph.commonNeighbors`, `SimpleGraph.mem_commonNeighbors`, `SimpleGraph.IsUniversal`; the instance `SimpleGraph.completeAtomicBooleanAlgebra`, through which the generic `compl_compl` (`Mathlib/Order/BooleanAlgebra/Basic.lean`) applies.
- `Mathlib/Combinatorics/SimpleGraph/Finite.lean`: `SimpleGraph.neighborFinset`, `SimpleGraph.card_neighborFinset_eq_degree`, `SimpleGraph.IsRegularOfDegree`, `IsRegularOfDegree.degree_eq`, `IsRegularOfDegree.compl`.
- `Mathlib/Combinatorics/SimpleGraph/Maps.lean`: `SimpleGraph.Iso` (notation $\simeq$`g`), `SimpleGraph.Iso.card_eq`, and `SimpleGraph.Embedding.complEquiv`; Mathlib has the complement of an embedding but no `Iso.compl`, which is new (a few lines, from `compl_adj` and the injectivity of `f`).
- `Mathlib/Order/SymmDiff.lean`: `symmDiff`, with the scoped notation $\triangle$ (`open symmDiff`); `Mathlib/Data/Finset/Card.lean`: `Finset.eq_of_subset_of_card_le`; `Mathlib/Data/Fintype/EquivFin.lean`: `Fintype.bijective_iff_injective_and_card` (Step 3); `Mathlib/Data/Finset/Powerset.lean`: `Finset.card_powersetCard` (the count $\binom{5}{2} = 10$).
- `Mathlib/Combinatorics/SimpleGraph/Triangle/Basic.lean`: `SimpleGraph.LocallyLinear`, which the triangle property implies (§10, row Lemma 2.8); it is not needed here.

7. Remarks on the sources

The direct argument for SE2 is Cameron's [C17, Theorem 2.2], and Seidel sketches a matrix proof [S68, §6, case 5]; we have written the direct argument in full. [BvM] §10.7 and §10.10 state both uniqueness theorems without proof. [BCN] §3.11 (iii) prints the parameters of the complement of the Clebsch graph as $(16, 5, 0, 1)$; they are $(16, 5, 0, 2)$, as `cgss/prerequisites.md` (SE2) says. Where `cgss/prerequisites.md` (SE3) says that the graph is unique "by GQ2", we use instead the repaired uniqueness of Lemma 2.10, as GQ3 allows: the triangle property of Lemma 4.1 is exactly what Proposition 4.3 needs, and the lines of the quadrangle never appear.

References

[BCN] A. E. Brouwer, A. M. Cohen and A. Neumaier, *Distance-regular graphs*, Springer, 1989.
[BvM] A. E. Brouwer and H. Van Maldeghem, *Strongly regular graphs*, Cambridge University Press, 2022.
[C17] P. J. Cameron, *Eigenvalues and root systems in finite geometry*, lecture notes, 2017.
[CGSS] P. J. Cameron, J.-M. Goethals, J. J. Seidel and E. E. Shult, Line graphs, root systems, and elliptic geometry, *J. Algebra* **43** (1976), 305–327.
[S68] J. J. Seidel, Strongly regular graphs with $(-1, 1, 0)$ adjacency matrix having eigenvalue 3, *Linear Algebra Appl.* **1** (1968), 281–298.